

Def.

On an interval I , the differential equation of the form

$$L(y) = a_n \cdot y^{(n)}(t) + a_{n-1} \cdot y^{(n-1)}(t) + \dots + a_1 y'(t) + a_0 y(t) = b(t)$$

is called the linear equation. If $b(t) \equiv 0$, we say that homogeneous.

Observation.

The L is a differential operator on $C^{(n)}$, because: for any $c \in \mathbb{R}$ and $y_1, y_2 \in C^{(n)}$,

$$L(cy_1 + y_2) = \sum_{i=0}^n a_i \cdot (cy_1 + y_2)^{(i)}(t) = \sum_{i=0}^n [c \cdot a_i y_1^{(i)}(t) + a_i y_2^{(i)}(t)] = c \cdot L(y_1) + L(y_2).$$

Order of 2.

Thm. Let $a_1, a_2 \in \mathbb{R}$ be constants, and consider the linear equation

$$L(y) = y'' + a_1 y' + a_2 y = 0.$$

If r_1, r_2 are distinct roots of the characteristic polynomial

$$p(r) = r^2 + a_1 r + a_2,$$

then $\phi(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ ($C_1, C_2 \in \mathbb{R}$)

is the general solution for $L(y) = 0$.

If r_1 is a repeated root of p , then

$$\phi(x) = C_1 e^{r_1 x} + C_2 x e^{r_1 x} \quad (C_1, C_2 \in \mathbb{R})$$

is the general solution.

Proof. observe that

$$L(e^{rx}) = r^2 \cdot e^{rx} + a_1 r e^{rx} + a_2 e^{rx} = (r^2 + a_1 r + a_2) \cdot e^{rx} = p(r) \cdot e^{rx}$$

and

$$L(x e^{rx}) = (x e^{rx})'' + a_1 (x e^{rx})' + a_2 (x e^{rx})$$

$$= (r e^{rx} + r e^{rx} + r^2 x e^{rx}) + a_1 e^{rx} + a_1 r x e^{rx} + a_2 x e^{rx}$$

$$= (r^2 + a_1 r + a_2) \cdot x e^{rx} + (2r + a_1) e^{rx}$$

$$= (p(r) \cdot x + p'(r)) \cdot e^{rx}$$

$p(r) = p'(r) = 0 \Leftrightarrow r$ is multiple root of p , so Proved.

Non-Homogeneous form.